Third Semester B. Tech. (Electronics and Communication Engineering) Examination

SIGNALS AND SYSTEMS

Time : 3 Hours] [Max. Marks : 60

Instructions to Candidates :—

(1) All questions are compulsory and carry marks as indicated.
(2) Assume suitable data wherever necessary.
(3) Illustrate your answers wherever necessary with the help of neat sketches.
(4) Due credit will be given to neatness.
(5) Use of Non – programmable calculator is permitted.

1. (a) Sketch the following Signals :

(i) $x(t) = r(t + 1) - 2r(t) + r(t - 1)$

(ii) $x(t) = \mathrm{r}(t) - 2r(t - 1) + 2r(t - 3) - r(t - 4)$

(iii) $x(n) = u(n + 2) - u(n - 3)$ 6(CO1)

(b) Compute the energy of the following signals :

(i) $x(t) = \cos(10\pi t)\ u(t)\ u(2 - t)$

(ii) $x(t) = r(t) - r(t - 2)$ 4(CO1)

2. (a) The system is represented in the form of differential equation :

$$\frac{d^3y(t)}{dt^3} + 4\frac{d^2y(t)}{dt^2} + 5\frac{dy(t)}{dt} + 2y^2(t) = x(t)$$

Check the system for :

(i) Linear or Non-Linear.

(ii) Causal or Non-Causal.

(iii) Time Invariant or Time Variant. 6(CO1)

(b) Elaborate the following properties of the system with example :

(i) Stability,

(ii) Causality,

(iii) Linearity,

(iv) Time-invariance. 4(CO1,4)

3. (a) The input and impulse response of DT-LTI system are
$x[n] = u[n+2] - 2u[n] + u[n-3]$;
$h[n] = 4\delta[n+2] + 3\delta[n+1] + 2\delta[n-1] + \delta[n-2] + \delta[n-3]$ respectively.
Obtain the response y[n] of the system. 6(CO3)

(b) The impulse responses of the LTI systems are given below. Verify the causality and stability of the system :

(i) $h_1[n] = \left[\frac{1}{5}\right]^n u[\mathrm{n}]$

(ii) $h_2[t] = e^{-6t}\, u(3-t)$ 4(CO3,4)

4. (a) Obtain the impulse response of the CT – LTI system if its output is $y(t) = e^{-t} - 2e^{-2t} + e^{-3t}$ and the input is $x(t) = e^{-0\cdot 5t}$. 6(CO2)

(b) CT – LTI system is described as –

$$\frac{d^2y(t)}{dt^2} - \frac{dy(t)}{dt} - 2y(t) = x(t)$$

Determine the impulse response $h(t)$ if the system is stable. 4(CO2)

5. (a) Find the Impulse response of the system described by the following difference equation using z-transform :

$$y[n] - 4y[n-1] + 3y[n-2] = x[n] + 2x[n-1]$$ 6(CO3)

(b) Obtain the inverse z-transform of $X(z) = \dfrac{z}{(z+2)(z-3)}$ for the Region of Convergence $2 < |z| < 3$. 4(CO3)

6. (a) Obtain the frequency response of the LTI system described by the following difference equation. Assume all initial conditions to be zero :

$$\frac{d^2y(t)}{dt^2} + 5\frac{dy(t)}{dt} + 6y(t) = 2x(t)$$

Plot the magnitude and phase spectrum of this system. 6(CO5)

(b) A DT LTI system is represented by

$$H(e^{j\omega}) = \frac{1}{1 + \frac{1}{2}e^{-j\omega} - \frac{1}{8}e^{-j2\omega}}.$$

Deduce the difference equation representing the input – output relationship of a DT system.

OR

Determine the Fourier Series representation for the signal :

$$x(t) = 4\cos\left(\frac{\pi}{2}t + \frac{\pi}{4}\right)$$

Plot its magnitude and phase spectrum. 4(CO5)